JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING & EXAM BRANCH

DEPARTMENT OF CIVIL ENGINEERING

★ SOLVED SEMESTER QUESTION PAPERS & MODEL ANSWERS ★

Strength of Materials (Mechanics of Solids) Semester PYQ with Solved Problems

Course Code:	R23-CIV-1202	**Academic Scheme:**	I B.Tech II Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Paper Section 1: Simple Stresses, Strains & Elastic Constants

Tensile, compressive and shear stresses, Stress-strain diagram for mild steel, Poisson's ratio, Volumetric strain, Relationships between E, G, and K, Thermal stresses.

Paper Section 2: Shear Force & Bending Moment Diagrams (SFD & BMD)

Cantilever, simply supported and overhanging beams subjected to point loads, UDL and UVL, Point of contraflexure determination.

Paper Section 3: Flexural & Shear Stresses in Beams

Bending equation M/I = sigma/y = E/R, Section modulus of rectangular, circular, I and T sections, Shear stress distribution across flanges and webs.

Paper Section 4: Deflection of Beams & Mohr's Theorems

Double integration method, Macaulay's method, Moment area method for slopes and deflections of cantilever and simply supported beams.

Paper Section 5: Torsion of Circular Shafts & Thin Cylinders

Torsion equation T/J = tau/r = G*theta/L, Polar modulus, Power transmitted, Stresses in thin cylindrical and spherical shells under internal pressure.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Max Marks	70 Marks (3 Hours)
Passing Criteria	40% in External (28/70) and 40% aggregate (40/100)
Model Paper Year	2023 - 2024 Academic Batches

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM